

Suresh Raja Selvadurai

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WORK EXPERIENCE

7+ years total experience

SelectDr

San Francisco, CA

Machine Learning Engineer | Recommender Systems · LLMs · FastAPI · React · AWS

Jan 2025 - Present

- Built **production healthcare recommendation systems** for specialty, condition, and procedure searches, ranking providers using clinical fit, quality, experience, distance, insurance, and patient signals.
- Shipped a **multi-agent LLM chatbot** with structured conversation state, retrieval, validation, safety routing, and FastAPI/React integration for guided healthcare search.
- Engineered **FastAPI APIs serving 1,000+ requests/second** and React workflows; built Databricks/PySpark, PostgreSQL/PostGIS, and OpenSearch pipelines over **100M+ healthcare records**.

Guardant Health

San Francisco, CA

Automation Software Intern | FastAPI · TCP · Docker · CUDA · Next.js

May 2024 - Aug 2024

- Built a Dockerized simulator and FastAPI control layer for laboratory robotics using **real-time TCP communication**, a Next.js GUI, Jenkins CI, and CUDA-backed performance optimization.

KPMG

Bangalore, India

Consultant, Data Science and AI | Client: Goldman Sachs | Java · Spring · Kubernetes · React

Nov 2020 - Aug 2023

- Built automated risk-management microservices and scalable internal APIs, contributing to workflows estimated to save **\$25M annually**.

Kissht

Mumbai, India

Data Scientist | XGBoost · scikit-learn · Credit Risk · A/B Testing

Dec 2019 - Nov 2020

- Developed **XGBoost credit-risk models** for high-volume lending; rebuilt post-COVID underwriting models to **0.65 AUC** and ran A/B tests to recalibrate decisions under distribution shift.

NameKart

Noida, India

Domain Name Analyst | Python · Selenium · Automation

Jul 2018 - Nov 2019

- Automated lead generation and contact discovery, increasing processing throughput from about 200 to **10,000 domains per day (50x)**.

EDUCATION

University of San Francisco

San Francisco, CA

M.S. in Computer Science | GPA: 3.95/4.00

Aug 2023 - May 2025

Indian Institute of Technology Patna

Patna, India

B.Tech. in Civil Engineering

Aug 2014 - Aug 2018

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, Java, TypeScript/JavaScript, C++, Rust
- Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, LLMs, RAG, recommender systems, NLP, model evaluation, A/B testing
- Data & Backend:** Databricks, PySpark, PostgreSQL, PostGIS, OpenSearch, H3, FastAPI, SQLAlchemy, Spring Boot
- Cloud & Infrastructure:** AWS, Docker, Kubernetes, Terraform, CI/CD
- Frontend:** React, Next.js

OTHER

Projects

RAG & Multimodal AI [LLMs · RAG · AWS SageMaker]

- Built retrieval-augmented and multimodal chatbot prototypes, including document-grounded question answering and multimodal inference workflows.

Systems & ML [C/C++/Rust · LSTM/RNN · Kotlin]

- Built an analytical-expression compiler, a stock-forecasting pipeline, and the Walnut personal-finance app.

Publications & Certifications

- Published research on *Compressibility Behavior of Bentonites*; Meta Front-End Development and IBM Machine Learning certifications.